

Using scientific data to show climate change in “Atlas of a Changing Earth”

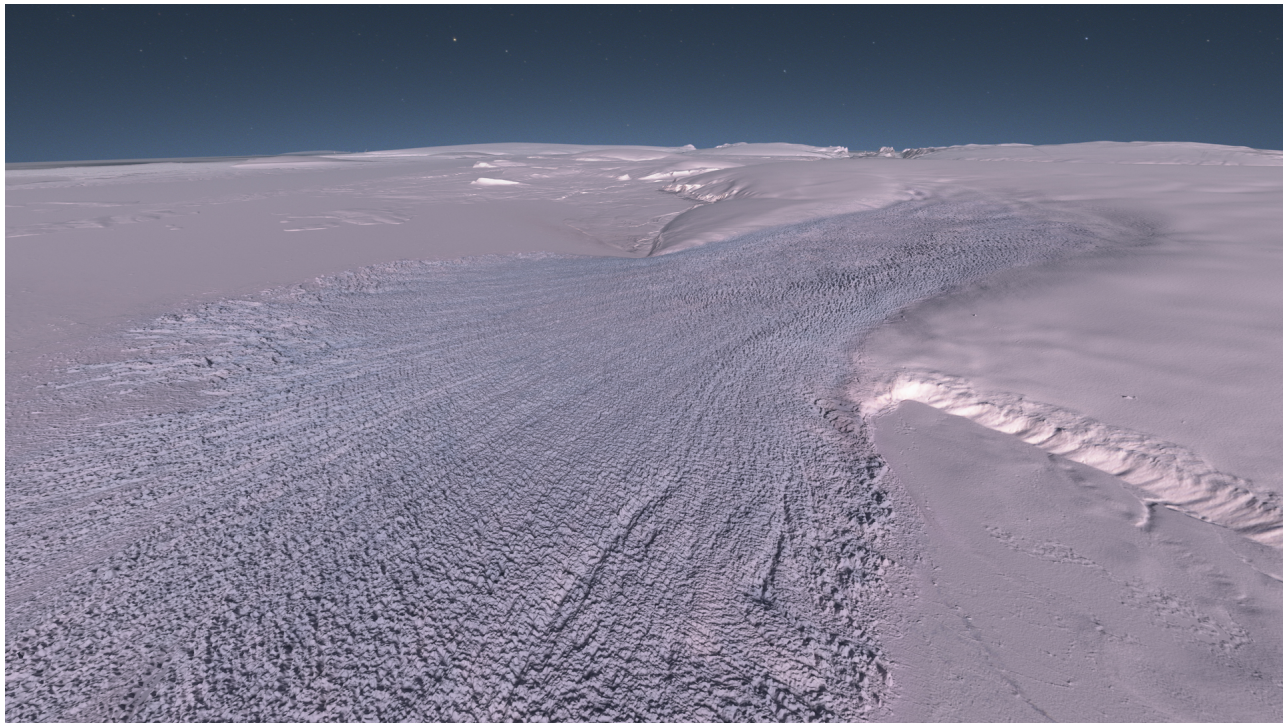
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ABSTRACT

The Advanced Visualization Lab (AVL) at the National Center for Supercomputing Applications created cinematic treatments of scientific data visualizations for the documentary film “Atlas of a Changing Earth”. This included the visualization of three types of data, each coming with numerous challenges: (1) arctic elevation models, (2) global climate simulation models, (3) African tree survey

data. In this session, we will discuss technical and design challenges associated in visualizing these datasets.

- (1) The ArcticDEM dataset is 200TB worth of elevation maps covering all land on Earth north of 60 degrees latitude, which took 1.5 billion hours of processing time on the Blue Waters supercomputer. The data, captured by satellite, is a goldmine of information, however comes with many challenges related to visualization: sporadic updates, misaligned image tiles, and cloud artifacts, to name a few. To create smooth-looking motion out of choppy infrequent data updates, we separated the images into low-frequency land and high-frequency ice passes, and performed motion flow and morphing on the high-frequency data. We manually cleaned cloud artifacts from the images and created a training set for a new machine learning algorithm called CloudFindr. We then combined the ArcticDEM data with numerous other datasets to tell the whole story – including landsat satellite images, an ocean ice simulation, NASA's Blue Marble Earth imagery, and a background of Hipparcos stars.
- (2) A global climate model simulation compared the year 2000 to a forecasted 2100 based on a “business as usual” emissions scenario. Visualizing this data involved both technical challenges like dealing with multiple coordinate systems, but also communication challenges in how best to represent the

data. Displaying temperature data directly did an insufficient job in telling the climate change story, as quick day-night temperature fluctuations made it difficult to notice patterns, and the viewer's eye was naturally drawn to areas of highest temperature absolute values rather than those with highest temperature changes, which is much more meaningful. Much data processing and iteration went into presenting the data in an understandable form.

- (3) A dataset of about 760 million individually-counted trees in Sub-Saharan Africa, identified using machine learning on satellite imagery. The visualization starts by looking at the African continent, and in one single seamless camera move goes down to a distance where individual trees can be identified. Each tree was represented with geometry of varying shapes in its correct location and correct size, the main challenge was in careful memory management and data representation such that different treatments could be seen at different scales.

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